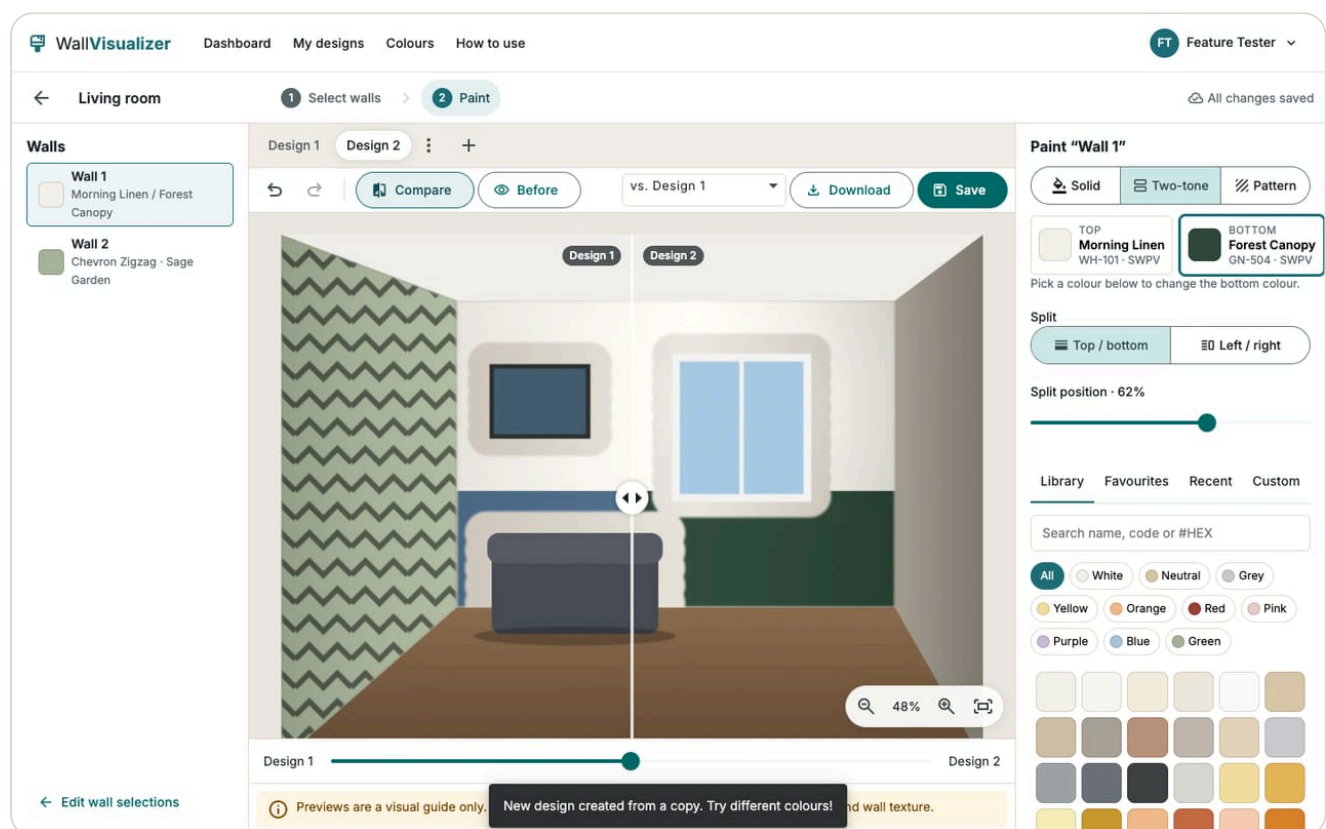


Smart Wall Paint Visualizer

A MEAN-stack web application that lets people preview paint colours, two-tone walls and patterns on a photo of their own room, with realistic light and shadow, before they buy paint.



Submitted by

Novshita

Programme

Unified Mentor internship

Technology

MongoDB · Express.js · Angular 22 · Node.js

Source code

github.com/novshita/smart-wall-paint-visualizer

Date

October 2026

Contents

1.	Introduction and problem statement
2.	Objectives and scope
3.	Users, roles and user flow
4.	System architecture
5.	Technology stack
6.	Database design
7.	Application pages
8.	Features in detail
9.	Realistic rendering method
10.	Security and privacy
11.	API design and documentation
12.	Admin panel and analytics
13.	Testing and quality assurance
14.	Deployment
15.	Challenges and how they were solved
16.	Results against the requirements
17.	Future scope
18.	Conclusion and learnings
19.	Appendix: running the project

Project at a glance

11 connected pages, all backed by the API	44 documented REST operations (Swagger)	194 automated tests (API, app and end-to-end)	~15 ms to recolour a wall (target: under 300 ms)
---	---	---	--

1. Introduction and problem statement

Choosing a wall colour is one of the most common home-improvement decisions, and one of the easiest to get wrong. A small paint chip under shop lighting rarely looks the same as a whole wall in a real room, with its own daylight, shadows and furniture. A wrong choice means buying more paint, repainting, and disappointment.

The project specification identifies four problems:

- Customers cannot accurately visualise the final painted look.
- Shade cards do not reflect real room lighting.
- Wrong choices cause dissatisfaction and repainting costs.
- No easy, user-friendly preview tool exists for ordinary users.

Smart Wall Paint Visualizer solves this with a browser-based tool. A user uploads a photo of their own room, marks the walls with simple tools, and tries colours, two-tone splits and patterns. The paint is blended so that the photo's own lighting, shadows and wall texture show through, which makes the preview look like paint rather than a flat coloured overlay. Designs can be saved, compared side by side, and downloaded at full resolution.

The same tool supports interior designers and painters during consultations, and paint retailers helping customers decide.

2. Objectives and scope

Objectives

PRIMARY OBJECTIVES	SECONDARY OBJECTIVES
Preview paint colours on the user's actual room photo	Support designers and painters in consultations
Reduce incorrect paint selection and repainting expense	Assist retailers in customer decision-making
Increase confidence in interior decisions	Encourage digital adoption in home-improvement planning
Provide a realistic, interactive visualisation experience	Demonstrate a practical MEAN-stack implementation

Scope (Phase 1)

IN SCOPE (BUILT)	OUT OF SCOPE
Room photo upload (JPG/PNG) with validation	Automatic AI wall detection
Manual wall selection: polygon, brush and eraser	AR/VR live camera preview
Solid, two-tone and pattern application	Native mobile application
Opacity, brightness and finish adjustment; before/after	Direct paint purchase and payment
Save, reopen, compare variants and download designs	
User and admin roles, admin dashboard with analytics	

3. Users, roles and user flow

ROLE	WHO	WHAT THEY CAN DO
Guest	Any visitor	Landing page, colour library, colour details, How to use guide
User	Homeowners, designers, painters, retail staff	Upload photos; select walls; paint, compare and save designs; download images; favourite colours; manage profile
Admin	System manager	Everything a user can do, plus manage colours, patterns, users and settings, view all designs and activity, and see the KPI dashboard

User flow

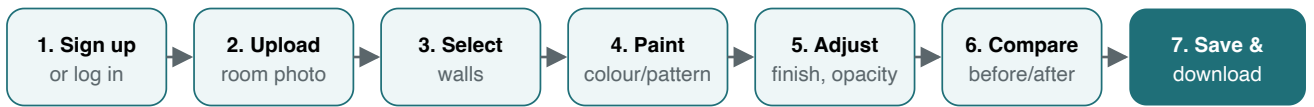


Figure 1. The seven-step user flow from the specification (§4), implemented end to end.

4. System architecture

The application follows a classic three-tier design: an Angular single-page app in the browser, a stateless Express REST API, and MongoDB for data. Images are kept out of the database in object storage and are only reachable through short-lived signed links.

Hosting: Vercel (app, proxies /api) → Render (API, Docker) → Atlas + S3

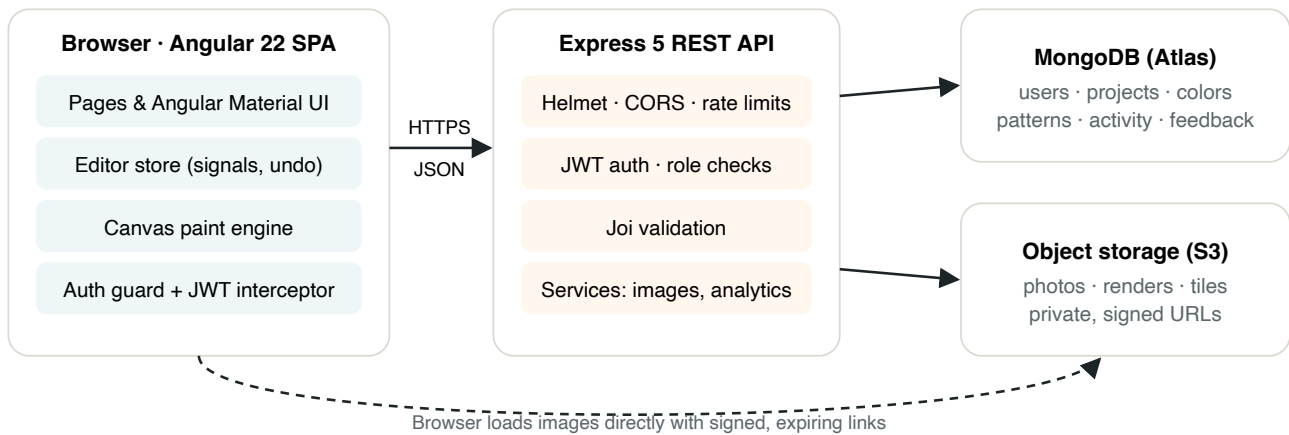


Figure 2. System architecture.

Design decisions

- **Rendering runs in the browser.** Recolouring happens on an HTML5 canvas in the user's browser, so changes appear instantly and the server never has to process images per click.
- **Walls are stored as vectors, not images.** A wall is a polygon plus brush/eraser strokes in coordinates normalised to the photo (0–1). This keeps saved designs small, fully editable, and re-renderable at any resolution.
- **Feature-based modular code.** The Angular app is split into lazy-loaded feature areas (upload, wall selection, studio, admin, ...) and the API into routes, controllers, services and middleware, as the specification's structure (§11) suggests.

5. Technology stack

LAYER	TECHNOLOGY	WHY
Frontend	Angular 22 (standalone components, signals), Angular Material, TypeScript, SCSS	Required MEAN stack; signals make the editor's state simple and fast; Material gives accessible, consistent components
Graphics	HTML5 Canvas, SVG	Pixel-level control for realistic blending; SVG for charts and diagrams
Backend	Node.js, Express 5, Joi, Mongoose 9	Lightweight REST API with validation on every endpoint
Database	MongoDB (Atlas in production)	Flexible documents fit nested designs (variants → walls → styles)
Images	sharp, AWS S3 (local disk in development)	Fast re-encoding, metadata stripping, resizing; private object storage

LAYER	TECHNOLOGY	WHY
Security	bcrypt, JWT, Helmet, express-rate-limit	Spec §12 requirements
Testing	Mocha + Supertest + in-memory MongoDB, Vitest, Playwright, axe	API, app, end-to-end and accessibility coverage
DevOps	Docker, GitHub Actions, Render, Vercel	Repeatable builds and automatic checks on every change

6. Database design

Seven MongoDB collections. Images are stored in object storage; the database keeps only their storage keys and metadata.

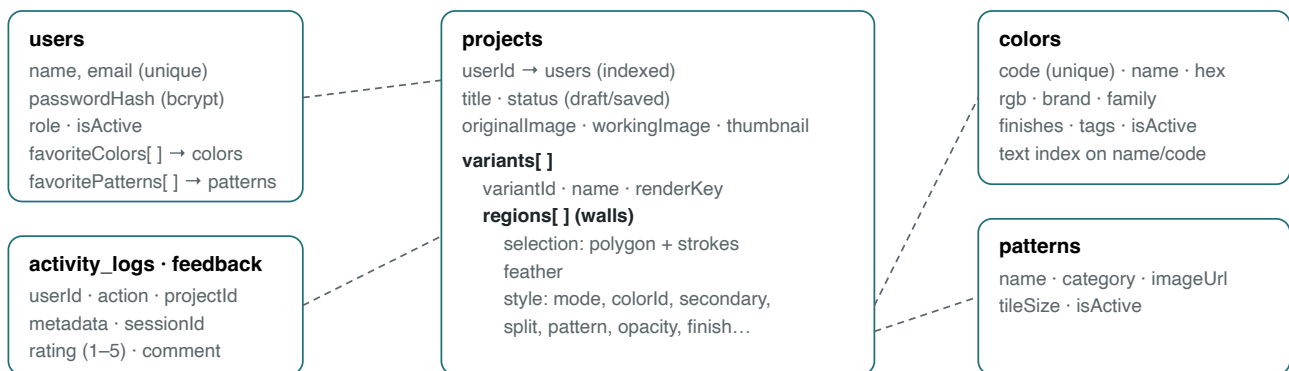


Figure 3. Main collections and relationships (a settings collection holds admin configuration).

COLLECTION	PURPOSE	KEY INDEXES
users	Accounts, roles, favourites, last login	email (unique)
projects	Photos, design variants, walls and styles	userId + updatedAt
colors	Paint catalogue (48 seed colours in 10 families)	code (unique), text (name, code), family, brand
patterns	Pattern tiles (8 seed patterns)	category
activity_logs	Logins, uploads, saves, downloads, sessions (for KPIs)	createdAt, action + createdAt
feedback	1–5 satisfaction ratings	—
settings	Upload limit, formats, default finish, disclaimer	key (unique)

7. Application pages

All eleven pages from the specification are built, connected through navigation, and backed by the API.

#	PAGE	ROUTE	ACCESS	PURPOSE
1	Landing	/	Public	Introduction, featured colours, call to action
2	Login / Register	/auth/login , /auth/register	Public	Authentication with field-level validation
3	Dashboard	/dashboard	User	Quick start, recent designs, favourite colours
4	Image upload	/projects/new	User	Drag-and-drop upload with checks and progress
5	Wall selection	/projects/:id/select	User	Polygon, brush and eraser tools
6	Studio	/projects/:id/studio	User	Paint, adjust, compare, save, download
7	Colour library	/colors	Public	Browse, search, filter, favourite
8	Saved designs	/projects	User	Reopen, rename, duplicate, download, delete
9	Profile	/profile	User	Account details, password, favourites
10	How to use	/help	Public	Tutorial, shortcuts, FAQ, disclaimer
11	Admin panel	/admin/*	Admin	Dashboard, colours, patterns, users, designs, activity, settings

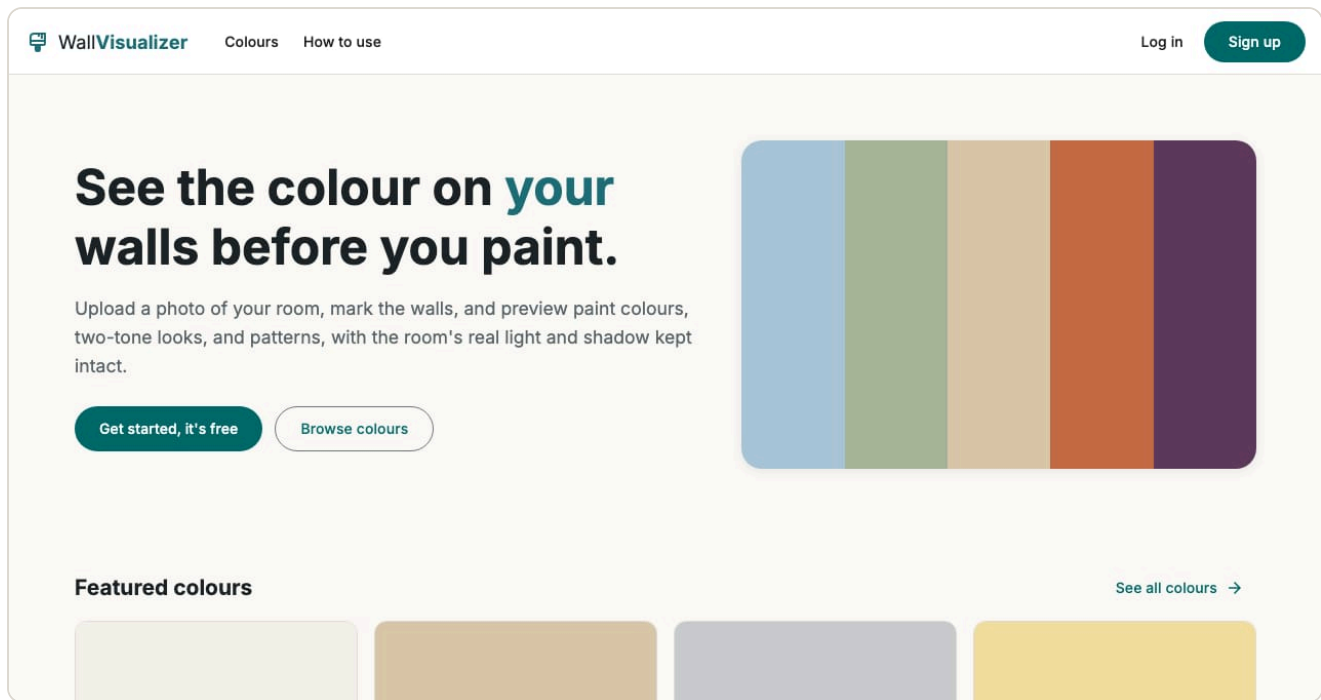


Figure 4. Landing page with featured colours from the live catalogue.

8. Features in detail

8.1 Uploading a room photo

Users drag in a photo or choose one. The browser checks the type, size and dimensions instantly and shows a preview; the server then checks everything again. Users must confirm that they own the photo, as the project guidelines require. On the server each upload is re-encoded, which removes hidden data such as GPS location, and stored in three sizes: the original (for full-quality downloads), a working copy of at most 2000 px (for fast editing), and a thumbnail.

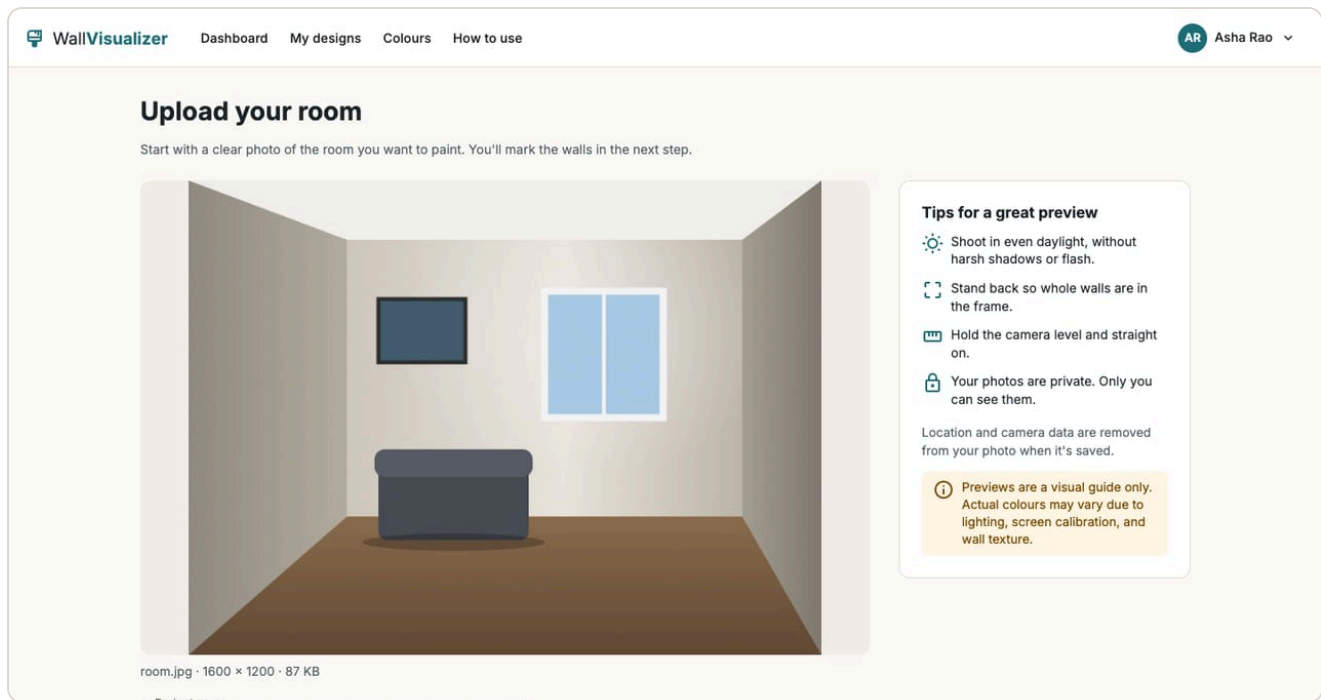


Figure 5. Upload page: preview, project name, ownership confirmation and photo tips.

8.2 Selecting walls

- **Polygon tool:** click each corner; close on the first point or press Enter; drag corners afterwards to adjust.
- **Brush and eraser:** paint to add awkward areas or remove windows, doors and furniture, with adjustable size.
- **Multiple named walls,** each painted independently; rename, delete with undo.
- **Precision aids:** zoom and pan (mouse wheel, pinch and two-finger pan on tablets), 100-step undo/redo, adjustable soft edges, keyboard shortcuts.
- A **first-use guide** explains the steps the first time someone opens the editor.

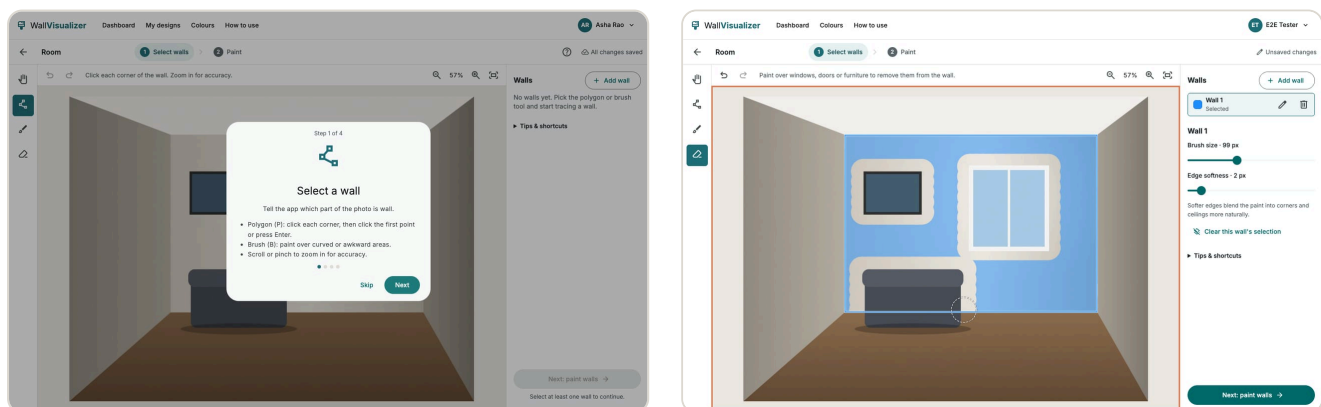


Figure 6. Left: the first-use guide. Right: the back wall traced with the polygon tool, with the window, picture and sofa removed with the eraser.

8.3 Painting in the studio

Each wall can be painted in one of three ways:

- **Solid colour** from the library, favourites, recently used colours, or any custom HEX code.
- **Two-tone**: two colours split horizontally or vertically at an adjustable position, like a painted dado line.
- **Pattern**: a tile repeated across the wall, with adjustable size and rotation and chosen background and pattern colours.

Finish (matte, satin, glossy), opacity, brightness and edge softness are adjustable per wall. The applied colour's name, code and brand are always shown, and a disclaimer reminds users that real results vary with lighting, screen calibration and wall texture.

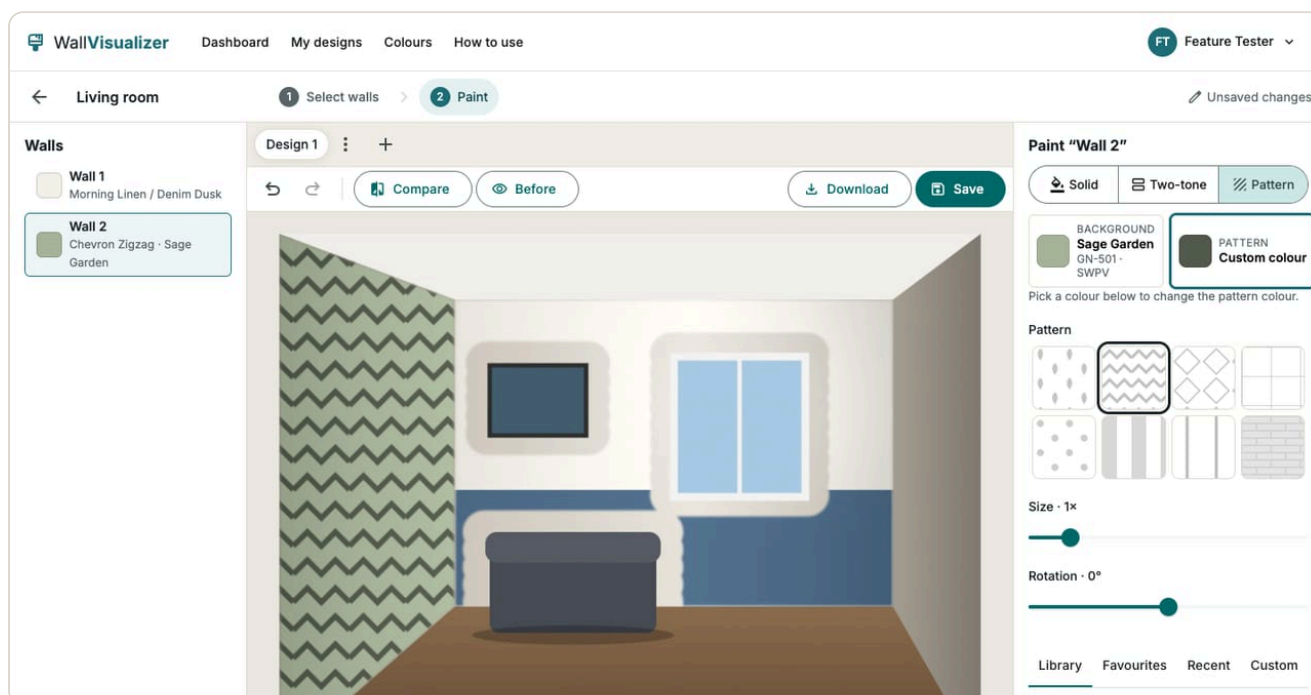


Figure 7. A two-tone back wall and a chevron-patterned side wall. The photo's lighting remains visible.

8.4 Design variants and comparison

Users can try several ideas on the same photo. A new variant copies the current colours; walls are shared across variants, so editing a wall's shape updates every design. The compare slider shows the original photo, or another design, on one side of a draggable divider.

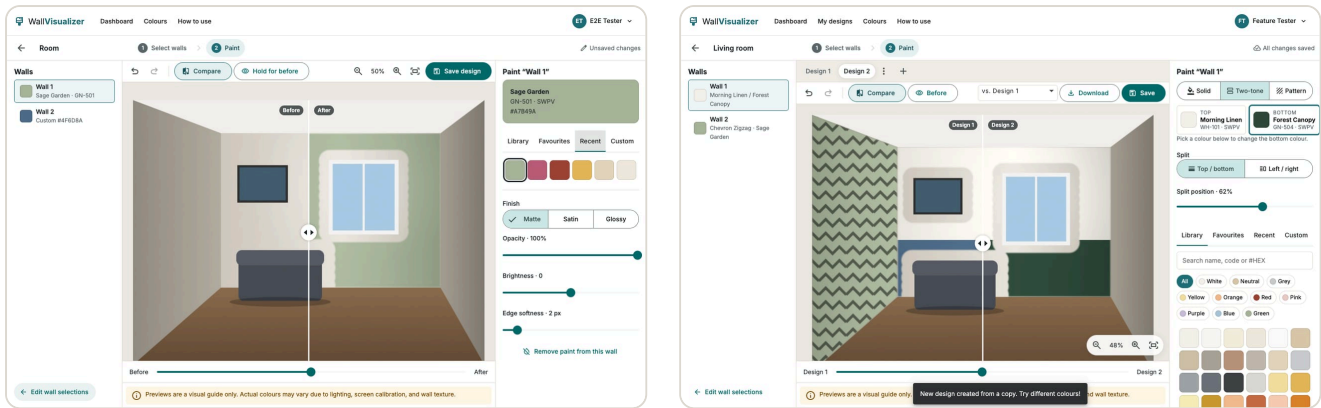


Figure 8. Left: before/after slider. Right: comparing "Design 1" with "Design 2".

8.5 Saving, downloading and managing designs

- **Auto-save** about one second after the user stops editing, with a visible status (Saving..., All changes saved, Retry).
- **Download** as PNG or JPG at the photo's full original resolution, either the painted room or a side-by-side before/after, with an optional caption listing each wall's colours and the disclaimer.
- **My designs** shows painted thumbnails with Draft/Saved status; designs can be reopened, renamed, duplicated (including their photos), downloaded and deleted.

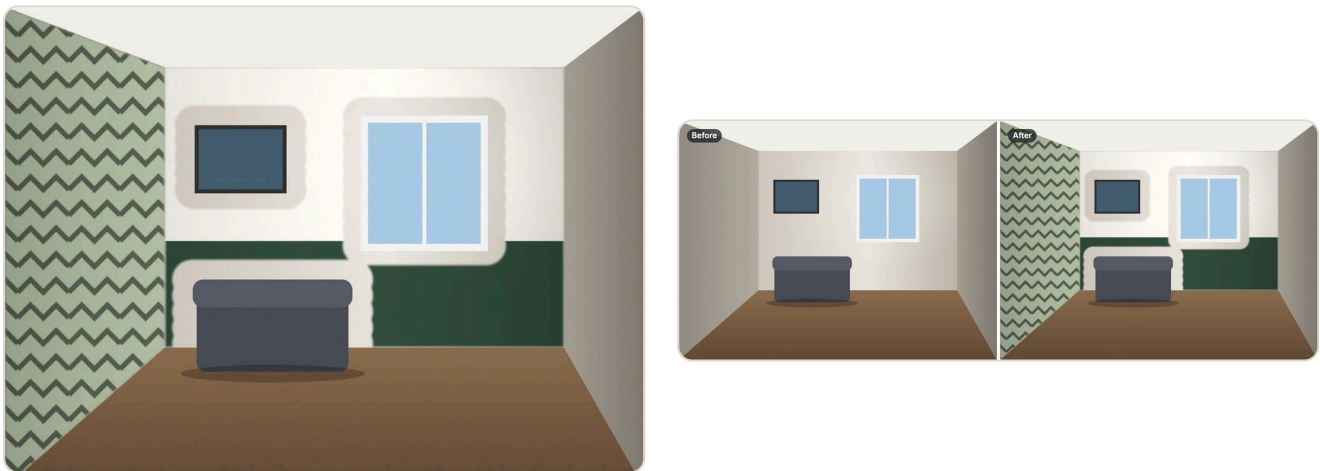


Figure 9. Real exports: painted room with colour caption (left) and a before/after image (right).

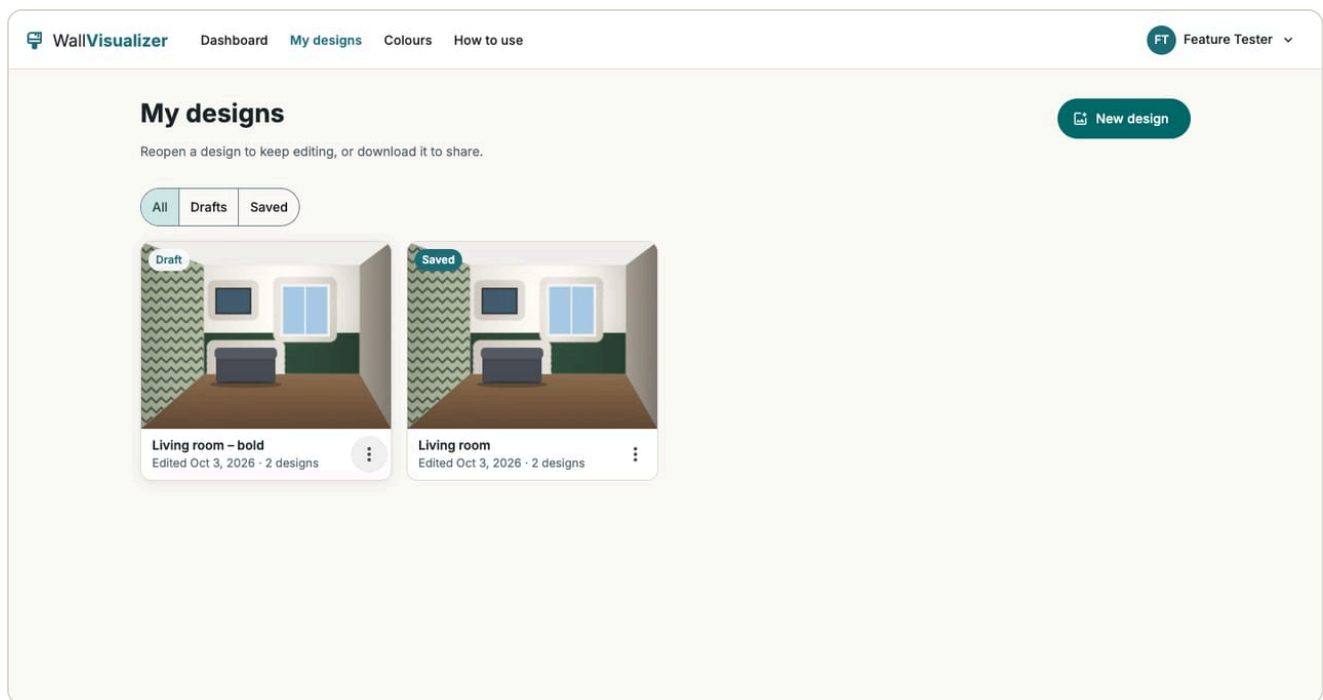


Figure 10. My designs: painted thumbnails, status badges and actions.

8.6 Colour library and favourites

The library groups colours by family and supports search by name, code or HEX, and filters by room and finish. Filters are kept in the page address, so a search can be bookmarked or shared. Each colour has a detail page with HEX/RGB values, finish previews, suitable rooms, related colours and a "Try on my room" button that starts an upload with that colour pre-selected.

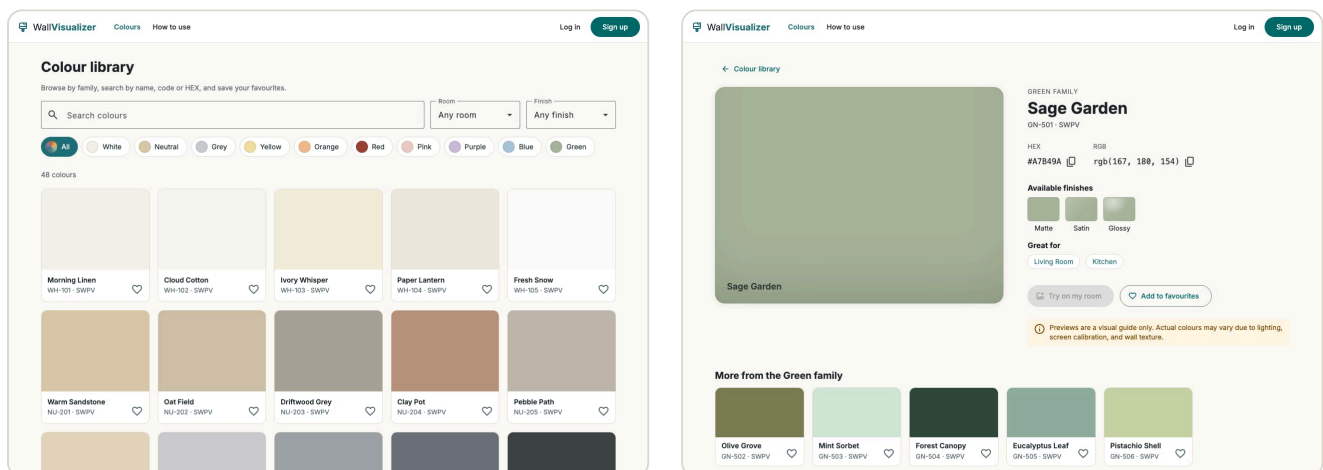


Figure 11. Colour library and colour detail page.

9. Realistic rendering method

The quality of the product depends on making painted walls look real. Simply drawing a semi-transparent colour over the photo looks flat and fake, because it hides the shading that tells the eye a

wall is three dimensional. The paint engine instead **keeps each pixel's shading and replaces only its colour**.

Algorithm

1. **Mask.** The wall's polygon and brush/eraser strokes are drawn into an alpha mask at the image's resolution. Optional feathering blurs the edge (three passes of a box blur, approximating a Gaussian).
2. **Luminance.** Every pixel's brightness is computed in *linear light* (not raw sRGB values), which is how light actually adds up.
3. **Relative shading.** The engine finds the wall's average brightness, then each pixel's shading s relative to that average: shadows have $s < 1$, highlights $s > 1$.
4. **Finish.** Highlights are flattened for matte ($\times 0.55$), kept for satin and boosted for glossy ($\times 1.4$).
5. **Recolour.** The paint colour is multiplied by the shading and brightness, converted back to sRGB, and mixed with the photo by the mask and opacity.

```
s = Lpixel / Lwall average  
resultlinear = paintlinear × finish(s) × 2brightness / 100  
output = mix(photo, toSRGB(result), mask × opacity)
```

A useful property: on a perfectly evenly lit wall the result is *exactly* the paint colour, while real photos keep their shadows, light falloff and texture. Two-tone walls choose the colour per pixel by the split line (with a 1-pixel anti-aliased edge); patterns mix the two colours by the tile's coverage.

Performance

Masks and luminance are cached, so changing a colour only re-runs the per-pixel blend, using lookup tables for the sRGB conversions. On a 1600 × 1200 photo a colour change takes about **11–21 ms** in Chrome, far inside the specification's target of 300 ms for a 2000 px image (§7). A unit test enforces the 300 ms budget on a 2000 × 1500 image.

Full-resolution export

Downloads re-run the same engine on the full-size original photo instead of the 2000 px working copy, as the specification requires (§8.6). Because walls are stored as vectors, they redraw sharply at any size.

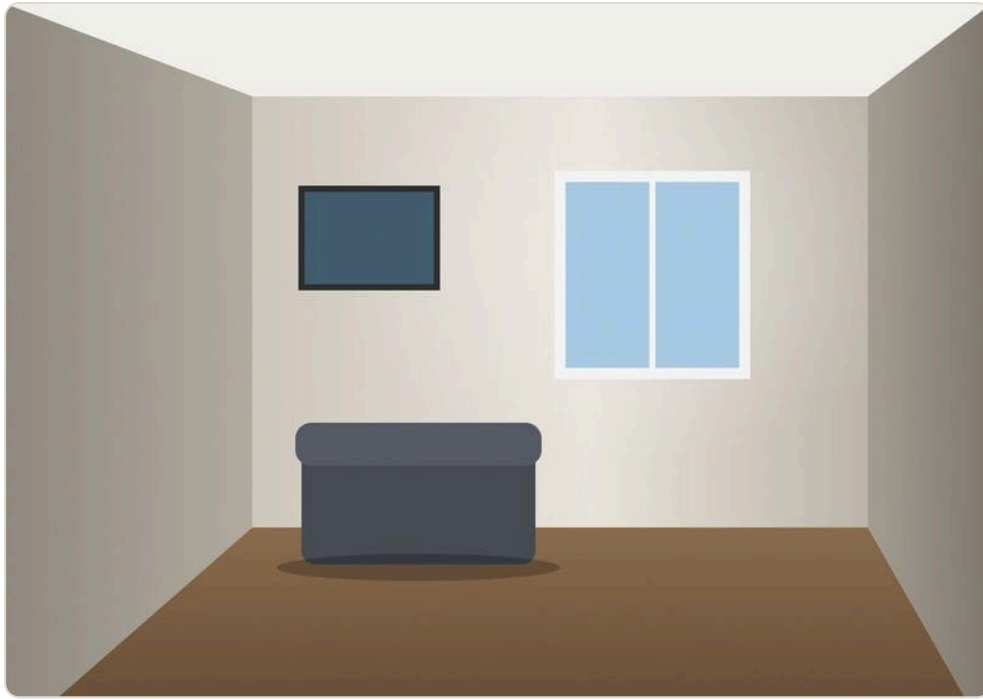


Figure 12. Original test photo (synthetic room, no copyrighted images). Compare with Figure 7.

10. Security and privacy

AREA	MEASURE
Passwords	bcrypt with cost 12; never returned by the API; strength rules (8+ characters, letter and number)
Sessions	JWT with 1-hour expiry; a password change or deactivation ends all existing sessions immediately (token versioning)
Login	Identical error and timing for wrong password vs unknown email, so attackers can't discover accounts; rate limiting
Authorisation	Role checks on every admin endpoint and page; users can only reach their own designs (others get "not found", so ids can't be probed); admins can't lock themselves out
Uploads	File type checked by content (magic bytes), not name; size limit set by admin; corrupt, tiny and oversized (50 MP+) images rejected; every image re-encoded to strip metadata and hidden payloads; SVG pattern uploads refused
Image privacy	Photos never served publicly; only short-lived signed links (HMAC locally, AWS Signature V4 presigned URLs on S3); deleting a design deletes its files
Input	Joi validation on every endpoint; NoSQL operator stripping; escaped search terms; size limits on request bodies
Transport & headers	HTTPS redirect and HSTS in production; Helmet headers; Content-Security-Policy; CORS allow-list

AREA	MEASURE
Dependencies	<code>npm audit</code> in CI: 0 known vulnerabilities at submission

Guidelines compliance. Users confirm photo ownership before uploading; the colour-accuracy disclaimer appears on the landing page, the studio, the help page and in exported images; all design, code and assets are original (colour names, pattern tiles and the test room photo were created for this project).

11. API design and documentation

The REST API lives under `/api/v1` and returns JSON, with one consistent error format (`{ status, message, errors[] }`) that the app uses to show field-level messages. All **44 operations** are documented with examples in an interactive **Swagger UI** at `/api/docs` , where reviewers can log in and try every endpoint.

GROUP	EXAMPLES
Auth	register, login, logout, get/update profile, change password
Projects	upload, list, get, save walls/styles, duplicate, upload render, delete
Catalogue	colours (list, search, facets, detail), patterns; admin create/update/remove and CSV/JSON import
Favourites	list, add and remove favourite colours and patterns
Admin	analytics, users, all designs, activity, settings
System	health check, public settings, activity events, feedback

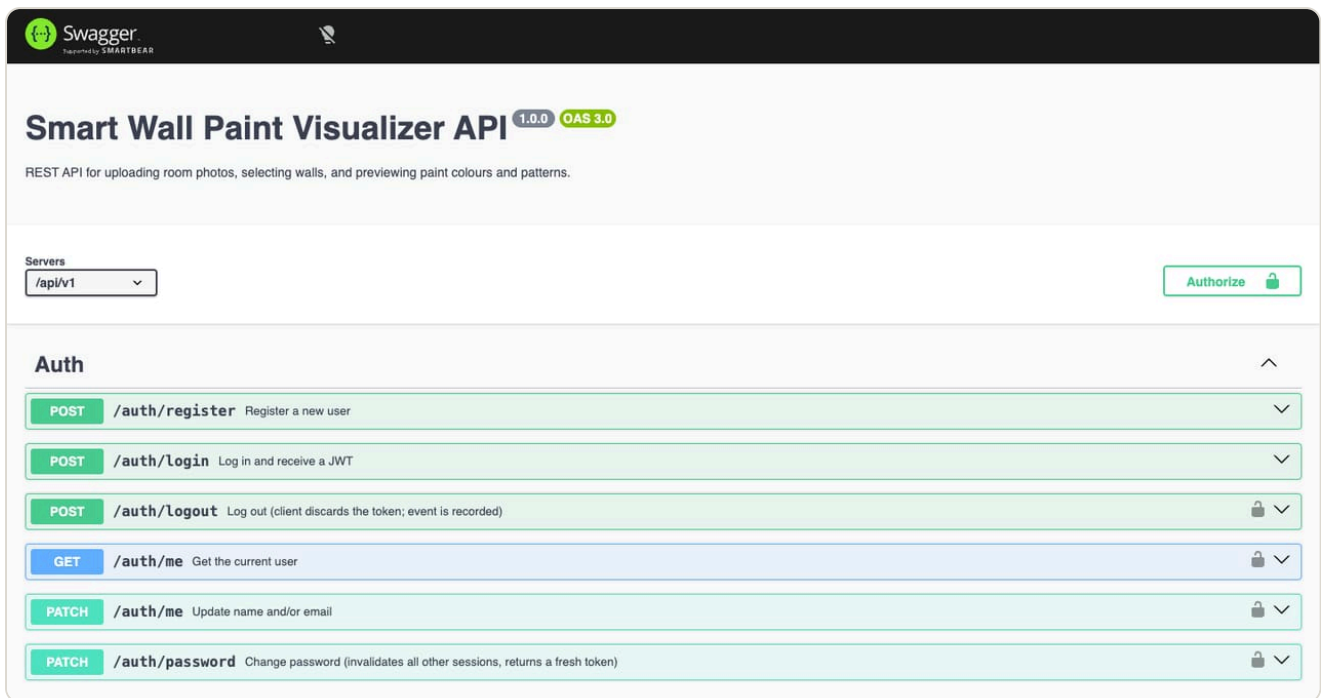


Figure 13. Interactive API documentation (Swagger UI).

12. Admin panel and analytics

- **Colours:** add, edit, remove and restore; bulk import from CSV or JSON with a per-row error report. Changes appear in the public library immediately.
- **Patterns:** upload tiles (converted to greyscale so users can tint them), edit and remove.
- **Users:** search, change roles, activate or deactivate.
- **Designs and activity:** read-only views of all users' designs and a filterable activity log.
- **Settings:** maximum upload size, allowed formats, default finish and the disclaimer text, applied immediately.

KPI dashboard

The dashboard shows the four KPIs from the specification (§14) for the last 7, 30 or 90 days and refreshes every 30 seconds: **room photos uploaded**, **designs saved**, **average session duration** (measured in the browser and sent even when the tab closes) and **user satisfaction** (a 1–5 star rating requested after downloads, at most once a week). It also shows daily trends with a table view, the rating distribution and the most-used colours. The chart colours were checked with a colour-vision-deficiency validator.

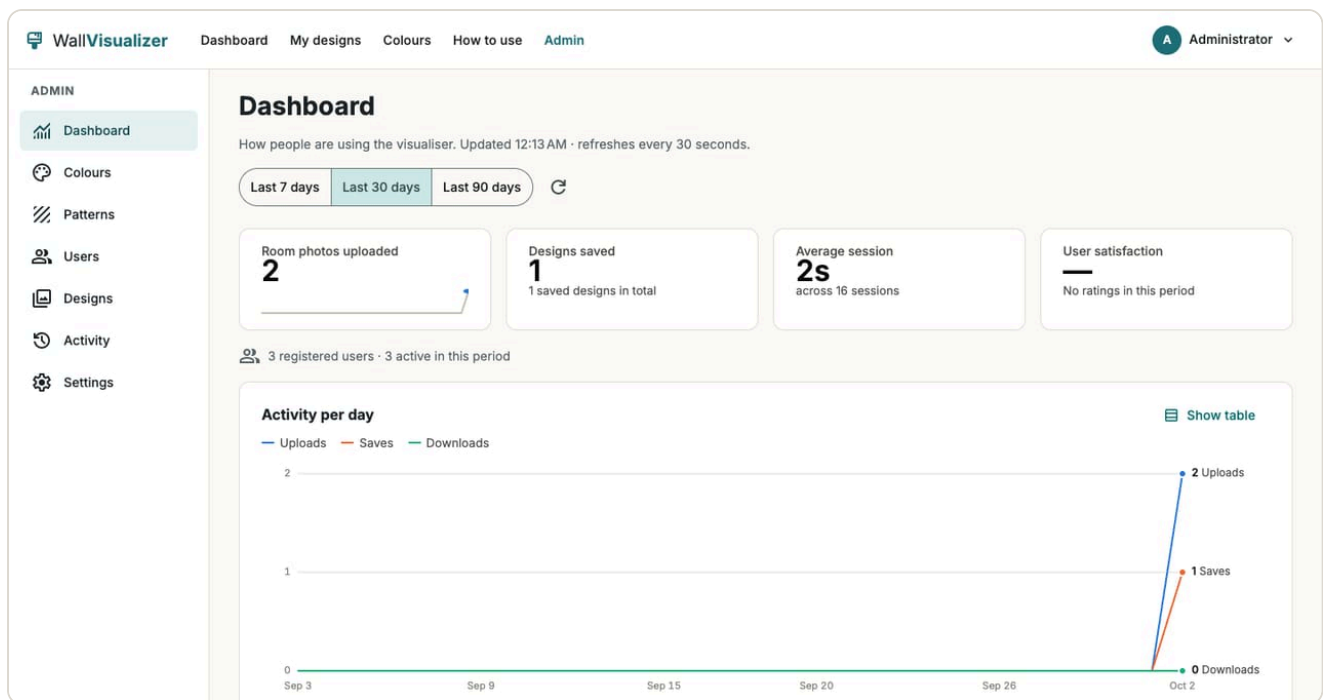


Figure 14. Admin KPI dashboard.

WallVisualizer Dashboard My designs Colours How to use Admin Administrator

ADMIN

- Dashboard
- Colours
- Patterns
- Users
- Designs
- Activity
- Settings

Colours

Changes appear in the colour library straight away.

Search name, code or HEX All colours

Import Add colour

COLOUR	CODE	FAMILY	BRAND	TAGS	STATUS
Heather Moor #8C7A9C	PU-954	Purple	SWPV	Living Room, Accent	Active
Lilac Morning #DCD0E6	PU-953	Purple	SWPV	Bedroom	Active
Plum Velvet #5E3B5C	PU-952	Purple	SWPV	Accent	Active
Lavender Haze #C9BCD9	PU-951	Purple	SWPV	Bedroom, Bathroom	Active
Raspberry Swirl #B85C78	PK-904	Pink	SWPV	Accent	Active
Ballet Slipper #F3DCDA	PK-903	Pink	SWPV	Bedroom	Active
Dusty Rose #C99A98	PK-902	Pink	SWPV	Bedroom, Living Room	Active
Blush Petal #EBCBC6	PK-901	Pink	SWPV	Bedroom	Active
Coral Reef #E7826D	RD-804	Red	SWPV	Living Room	Active

Figure 15. Colour management with add, edit, remove and import.

13. Testing and quality assurance

102

API tests (Mocha, Supertest, in-memory MongoDB)

89

app tests (Vitest)

3

end-to-end journeys (Playwright, real browser)

22

pages and dialogs passing WCAG 2.1 AA checks (axe)

LEVEL	WHAT IS COVERED
API	Authentication and sessions; upload validation (fake files, GPS stripping, orientation, size limits); access control; signed links (tampering, expiry, path traversal); designs; admin features; CSV import; analytics; AWS signing checked against AWS's published example
App	Auth service and guards; forms; colour library; upload flow; editor geometry, undo history, zoom maths; the paint blend (flat wall equals the target colour, shadows kept, opacity, finish, two-tone, patterns); render-time budget; editor store (auto-save, variant sync, undo grouping)
End-to-end	Register → upload → select with both tools → paint → compare → save → download → reopen → delete; admin KPIs and colour management; blocked access for guests and other users
Accessibility	Automated WCAG 2.1 AA audit of every page; keyboard navigation and shortcuts; visible focus; labels and alt text; screen-reader announcements
Visual	Scripted browser runs with screenshots at desktop (1440 px) and tablet (1024 px) sizes; exported files checked for resolution and content
Continuous integration	GitHub Actions on every change: lint → tests → security audit → build → end-to-end tests → deploy

Linting (ESLint) and formatting (Prettier) are enforced on both projects, and every feature was merged through a pull request with all checks passing.

14. Deployment

LAYER	SERVICE	NOTES
Web app	Vercel	HTTPS, security headers; forwards <code>/api</code> to the API so the browser uses one domain
API	Render (Docker)	Non-root container, health check, HTTPS redirect

LAYER	SERVICE	NOTES
Database	MongoDB Atlas	Managed, backed up
Images	AWS S3	Private bucket; presigned links; public prefix only for pattern tiles
CI/CD	GitHub Actions	Deploys automatically after all tests pass on <code>main</code>

Environment configuration is kept in environment variables, never in code. A step-by-step guide is in `docs/deployment.md`. The project is fully prepared for deployment (Docker image, hosting configuration and CI/CD pipeline); a public live instance was not set up for this submission, so the application is demonstrated by running it locally (see the appendix).

15. Challenges and how they were solved

CHALLENGE	SOLUTION
Painted walls looked flat and fake with simple colour overlays	Designed a luminance-preserving blend in linear light that keeps each pixel's shading relative to the wall's average (Section 9)
Rendering speed on large photos	Edit on a 2000 px working copy; cache masks and luminance; lookup tables; coalesce redraws to one per frame; export separately at full resolution
Saving brush selections efficiently and keeping them editable	Stored walls as vector polygons and strokes in normalised coordinates instead of mask images
The test framework (Jest) could not connect to the test database on Node 26	Diagnosed a driver handshake problem inside Jest's sandbox and switched API tests to Mocha
Seeded colours were missing RGB values, crashing the colour detail page	Found through screenshot testing; fixed the seed script, backfilled existing data, and added tests for seed data
The first corner of a polygon was sometimes lost	Browser testing revealed a timing issue; tied unfinished polygons to their wall so behaviour no longer depends on timing
Pressing Enter to finish a polygon also clicked the last-used button	Made the canvas take keyboard focus when clicked
Signed image links changed on every request, so browsers re-downloaded images	Made links stable within each hour so they can be cached, while still expiring
Accessibility issues in the wall list and variants bar (nested buttons, incomplete tab roles)	Restructured both components; all pages now pass the automated WCAG 2.1 AA audit

CHALLENGE	SOLUTION
End-to-end tests failed only in CI ("port already in use")	Traced it to a shared PORT variable read by both servers; pinned each server's port explicitly

16. Results against the requirements

ACCEPTANCE CRITERION (SPEC §20)	RESULT
New user can register, upload, select a wall with both tools, apply a colour, compare, save and download using only the in-app guide	Met (automated end-to-end test)
Painted walls retain visible shadows and texture	Met (Figures 7–9; unit tests)
Dual-tone and patterns on at least one region; multiple regions per image	Met
Saved projects reopen with selections and styles fully editable	Met
Admin can add, edit and remove colours and patterns, reflected immediately	Met (end-to-end test)
Admin dashboard displays all four KPIs	Met (end-to-end test)
Unauthorised access to others' projects and admin routes is blocked	Met (API and end-to-end tests)
At least 6–7 interconnected pages integrated with backend and database	Met (11 pages)
Recolour under 300 ms for a 2000 px image	Met (~15 ms)
Live on a cloud environment with HTTPS	Deployment-ready (Docker, hosting configuration, CI/CD and guide); not hosted live

17. Future scope

- **AI wall detection:** automatic wall segmentation to replace manual selection, with the existing tools for corrections.
- **Lighting simulation:** day, evening and warm/cool light previews.
- **Brand integration:** catalogue sync with paint manufacturers, "buy this colour" links and quantity estimates from wall area.

- **Mobile and AR:** a native app and live camera preview.
- **Sharing and collaboration:** share a read-only design link with a client or family member.
- **Performance:** move pixel processing to a Web Worker for very large photos on low-end devices.

18. Conclusion and learnings

Smart Wall Paint Visualizer meets the Phase 1 specification as a complete, end-to-end MEAN application: eleven connected pages, a realistic browser-side rendering engine, secure handling of private photos, an admin panel with live KPIs, and 194 automated tests running on every change.

Key learnings

- **Image processing fundamentals:** colour spaces, linear light and luminance turned out to be the key to realistic results.
- **Building for real users:** undo, auto-save, clear errors, a first-use guide and accessibility matter as much as the core feature.
- **Security by design:** validating files by content, signing private links and enforcing ownership on every request.
- **Testing in a real browser finds real bugs:** several issues (missing data, timing and keyboard focus problems) only appeared in end-to-end runs and were fixed before release.
- **Working in small, reviewed steps:** each feature was built on its own branch and merged through a pull request with automated checks.

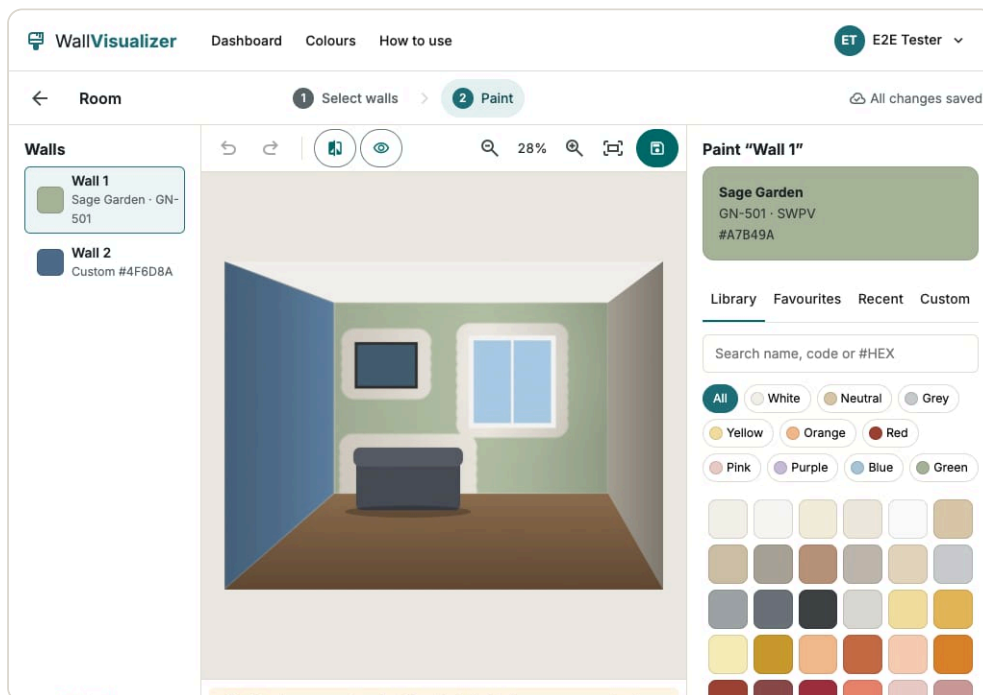


Figure 16. The studio on a tablet-sized screen (1024 px).

19. Appendix: running the project

Requirements

Node.js 18 or later, Docker (for a local MongoDB) and Git.

Steps

```
git clone https://github.com/novshita/smart-wall-paint-visualizer.git
cd smart-wall-paint-visualizer
docker compose up -d          # MongoDB

cd server && npm install && cp .env.example .env
npm run seed                  # colours, patterns, admin account
npm run dev                   # API on http://localhost:5050

cd ../client && npm install && npm start # app on http://localhost:4200
```

Accounts and documentation

- **Local admin account** (created by the seed): `admin@swpv.local` / `ChangeMe123!`.
- **API documentation:** `http://localhost:5050/api/docs`
- **Tests:** `npm test` in `server`; `npx ng test` in `client`; `npx playwright test` in `e2e`.
- **Specification:** `docs/spec.md` · **Deployment guide:** `docs/deployment.md`